Quantitative Research – Mock Exam 1

Discussion of the solutions

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Contents

1. Task 1: Statistical inference: z-test and confidence interval
2. Task 2: Simple and multiple linear regression
3. Task 3: Classification: logistic regression and evaluation

The mock exam at a glance

Task	Topic	Scenario	Points
1	Statistical inference: z-test and confidence interval	Order values of an online shop	20
2	Simple and multiple linear regression	Temperature and iced-coffee sales of a café	20
3	Classification: logistic regression and evaluation	Churn prediction of a mobile-phone provider	20
Total	3×10 subtasks worth 2 points each		**60**

How the exam is built

Every subtask can be solved **independently** and asks for exactly **one** result. No intermediate result is needed in a later subtask – all values required are stated in the subtask itself.

Overview

1. Task 1: Statistical inference: z-test and confidence interval
2. Task 2: Simple and multiple linear regression
3. Task 3: Classification: logistic regression and evaluation

Task 1 – scenario

Order values after the website redesign

Planning figure $\mu_0 = 80$ EUR, known $\sigma = 12$ EUR

Random sample: $n = 36$ orders, $\bar{x} = 84.30$ EUR, $\alpha = 0.05$

- Question: has the mean order value *changed* (in either direction)? $\Rightarrow$ **two-sided** test.
- σ known $\Rightarrow$ Z_n and the Φ-table; $\sqrt{36} = 6$ keeps the numbers clean.

Task 1 – the concepts you need

- **Standard error** (σ known): $\mathrm{SE} = \sigma/\sqrt{n}$ – how strongly $\bar{X}$ fluctuates around μ.
- **Test statistic:** $Z_n = \dfrac{\bar{X} - \mu_0}{\sigma/\sqrt{n}} \sim N(0,1)$ under H_0; two-sided: reject H_0 if $|Z_n| > z_{1-\alpha/2}$.
- p**-value** (two-sided): $p = 2\,\Phi(-|Z_n|)$; reject H_0 if $p < \alpha$.
- **Confidence interval:** $\mathrm{CI}_{1-\alpha} = \left[\bar{X} \pm z_{1-\alpha/2}\,\sigma/\sqrt{n}\right]$; μ_0 outside the CI $\Leftrightarrow$ reject H_0 (duality).
- **Type I error:** reject H_0 although H_0 is true (false alarm); its probability is α.

Task 1 – the test step by step

Step	Formula	Numbers	Value
Hypotheses	$H_0: \mu = \mu_0$ vs. $H_1: \mu \neq \mu_0$	$\mu_0 = 80$	two-sided
Standard error	$\sigma/\sqrt{n}$	$12/\sqrt{36}$	2
Test statistic	$Z_n = (\bar{x} - \mu_0)/(\sigma/\sqrt{n})$	$(84.30 - 80)/2$	2.15
Critical value	$z_{1-\alpha/2}$	$z_{0.975}$	1.96
Decision	$\lvert Z_n \rvert > z_{1-\alpha/2}$?	$2.15 > 1.96$	reject H_0
p-value	$2\,\Phi(-\lvert Z_n \rvert)$	$2\,(1 - 0.9842)$	0.0316
95 % CI	$\bar{x} \pm z_{0.975}\,\sigma/\sqrt{n}$	84.30 ± 3.92	[80.38; 88.22]

Task 1 – three routes, one verdict

Critical value	p-value	Confidence interval
$\lvert Z_n \rvert = 2.15$	$p = 2\,(1 - \Phi(2.15))$	[80.38; 88.22]
$z_{0.975} = 1.96$	$= 0.0316$	$\mu_0 = 80$ outside
$2.15 > 1.96$	$0.0316 < 0.05$	
$\Rightarrow$ **reject** H_0	$\Rightarrow$ **reject** H_0	$\Rightarrow$ **reject** H_0

- Type I error here: "the mean changed" although it is still 80 EUR – probability $\alpha = 0.05$.
- σ unknown in practice: `stats.ttest_1samp(orders, popmean=80)` (two-sided by default, t_{35}).

Key takeaway

Critical value, p-value and CI are three views of the same test – they always agree.

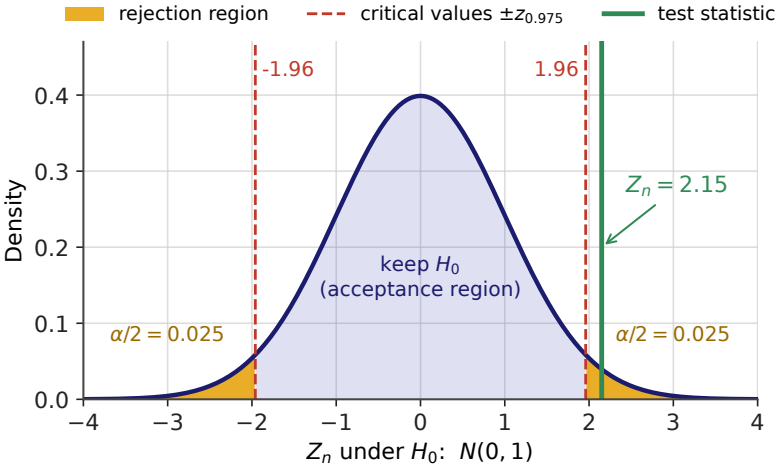
Task 1 – results a)–e)

Part	Asked for	Result
a)	Hypothesis pair	$H_0: \mu = 80$ vs. $H_1: \mu \neq 80$
b)	Standard error	$\sigma/\sqrt{n} = 12/6 = 2$ EUR
c)	Test statistic Z_n	$Z_n = 4.30/2 = 2.15$
d)	Critical value	$z_{1-\alpha/2} = z_{0.975} = 1.96$
e)	Test decision	$\lvert Z_n \rvert = 2.15 > 1.96 \Rightarrow$ reject H_0

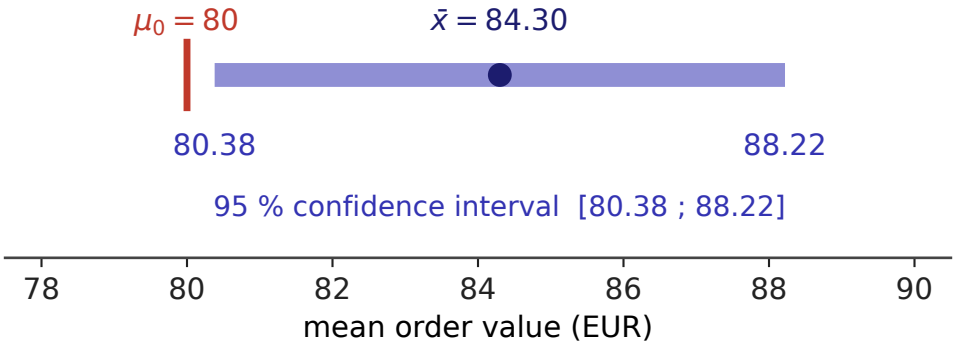
Task 1 – results f)–j)

Part	Asked for	Result
f)	p-value	$p = 2\,(1 - 0.9842) = 0.0316$
g)	95 % confidence interval	$84.30 \pm 3.92 = [80.38;\ 88.22]$ EUR
h)	Interpretation of the CI	with 95 % confidence, the mean μ of *all* orders lies in $[80.38;\ 88.22]$ EUR
i)	Type I error in context	rejecting H_0 ("mean changed") although $\mu = 80$ EUR still holds
j)	Python command (t-test)	`stats.ttest_1samp(orders, popmean=80)`

Task 1 – rejection region and test statistic



Task 1 – confidence interval and duality



Task 1 – typical mistakes

- Using the one-sided quantile $z_{0.95} = 1.645$ for the two-sided test.
- Forgetting the factor 2 in the two-sided p-value (0.0158 instead of 0.0316).
- Standard error with n or $n - 1$ instead of $\sqrt{n}$.
- Reading the CI as the range of single order values, or as "95 % probability for *this* interval".
- Mixing up the Type I error (false alarm: reject a true H_0) and the Type II error (miss).

Overview

1. Task 1: Statistical inference: z-test and confidence interval
2. Task 2: Simple and multiple linear regression
3. Task 3: Classification: logistic regression and evaluation

Task 2 – scenario

Part A: five summer days

x_i (°C)	20	22	24	26	28
y_i	46	54	62	70	68

$\bar{x} = 24$, $\bar{y} = 60$

$S_{xx} = 40$, $S_{xy} = 120$, $S_{yy} = 400$

Part B: 60 days, two explanatory variables

`sales` $\sim$ `temp + rain`

	coef	P>|t|
`temp`	3.0055	0.000
`rain`	−0.4313	0.253

Task 2 – the concepts you need

- **OLS estimators:** $\hat{\beta}_1 = S_{xy}/S_{xx}$, $\hat{\beta}_0 = \bar{y} - \hat{\beta}_1\bar{x}$; line $\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x$.
- **Residual:** $\hat{u}_i = y_i - \hat{y}_i$ – positive: point above the line, negative: below.
- **Sum-of-squares decomposition:** $\text{TSS} = \text{ESS} + \text{RSS}$, $R^2 = \text{ESS}/\text{TSS} = 1 - \text{RSS}/\text{TSS}$.
- **Correlation** (simple regression): $R^2 = r_{xy}^2$, so $r_{xy} = \pm\sqrt{R^2}$ with the sign of $\hat{\beta}_1$.
- **Multiple regression:** $\hat{\beta}_j$ = change of y per unit of x_j *holding the other variables fixed*; $H_0\colon \beta_j = 0$ is rejected if $p < \alpha$.

Task 2 – line, fitted values and residuals

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{120}{40} = 3, \qquad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 60 - 3 \cdot 24 = -12 \qquad \Rightarrow \qquad \hat{y} = -12 + 3x$$

x_i	20	22	24	26	28	$\sum$
y_i	46	54	62	70	68	300
$\hat{y}_i$	48	54	60	66	72	300
$\hat{u}_i = y_i - \hat{y}_i$	-2	0	$+2$	$+4$	-4	0
$\hat{u}_i^2$	4	0	4	16	16	$40 = \text{RSS}$

- Prediction at $x = 25$ (inside the data range): $\hat{y} = -12 + 75 = 63$.
- Slope: $1\,^\circ$C warmer $\Rightarrow$ on average 3 more iced coffees; the intercept -12 is pure extrapolation.

Task 2 – sums of squares, R^2 and r_{xy}

$$\underbrace{\text{TSS}}_{S_{yy}=400} = \underbrace{\text{ESS}}_{\hat{\beta}_1^2 S_{xx}=9\cdot 40=360} + \underbrace{\text{RSS}}_{\sum \hat{u}_i^2=40}$$

$$R^2 = \frac{\text{ESS}}{\text{TSS}} = 1 - \frac{\text{RSS}}{\text{TSS}} = 1 - \frac{40}{400} = 0.9, \qquad r_{xy} = +\sqrt{0.9} = 0.9487$$

- 90 % of the variation in sales is explained (linearly) by the temperature.
- The sign of r_{xy} comes from $\hat{\beta}_1 > 0$; check: $120/\sqrt{40 \cdot 400} = 0.9487$.

Key takeaway

$R^2 = r_{xy}^2$ holds in *simple* regression only – take the root and add the sign of the slope.

Task 2 – reading the multiple regression output

```
Dep. Variable:                    sales   R-squared:                       0.947
Model:                              OLS   Adj. R-squared:                  0.945
Method:                   Least Squares   F-statistic:                     512.4
                 coef    std err          t      P>|t|      [0.025      0.975]
------------------------------------------------------------------------------
Intercept    -12.2152      3.783     -3.229      0.002     -19.790      -4.641
temp           3.0055      0.131     22.863      0.000       2.742       3.269
rain          -0.4313      0.374     -1.154      0.253      -1.180       0.317
```

- `temp`: holding `rain` fixed, $+1\,^\circ\text{C} \Rightarrow$ on average $+3.01$ iced coffees ($p = 0.000$).
- `rain`: $p = 0.253 > 0.05 \Rightarrow$ keep $H_0\colon \beta_2 = 0$ – not significant *given the temperature*.
- Confounding: alone, `sales` $\sim$ `rain` has slope -6.19 – rainy days are also cooler days.

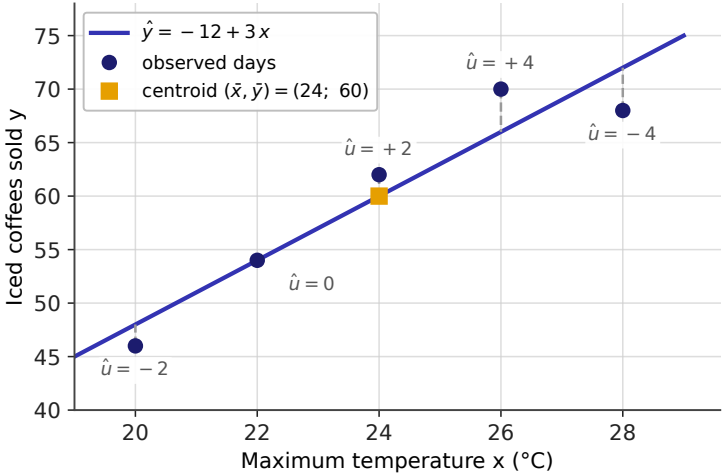
Task 2 – results a)–e)

Part	Asked for	Result
a)	Slope $\hat{\beta}_1$	$\hat{\beta}_1 = 120/40 = 3$
b)	Intercept $\hat{\beta}_0$	$\hat{\beta}_0 = 60 - 3 \cdot 24 = -12$
c)	Interpretation of the slope	$1\,^\circ$C warmer $\Rightarrow$ on average about 3 more iced coffees sold
d)	Prediction at $x = 25\,^\circ$C	$\hat{y}(25) = -12 + 75 = 63$ iced coffees
e)	Residual of day 5	$\hat{u}_5 = 68 - 72 = -4$

Task 2 – results f)–j)

Part	Asked for	Result
f)	Coefficient of determination R^2	$R^2 = 1 - 40/400 = 0.9$
g)	Correlation coefficient r_{xy}	$r_{xy} = +\sqrt{0.9} = 0.9487$
h)	Python command (OLS)	`smf.ols("sales ~ temp", data=cafe).fit()`
i)	Ceteris-paribus interpretation	holding rain fixed: $+1\,°C \Rightarrow$ on average 3.01 more iced coffees
j)	Significance of `rain`	$p = 0.253 > 0.05 \Rightarrow$ not significant (keep H_0: $\beta_2 = 0$)

Task 2 – scatter plot, OLS line and residuals



Task 2 – typical mistakes

- $\hat{\beta}_1 = S_{xx}/S_{xy}$ instead of S_{xy}/S_{xx}.
- Residual with the wrong sign ($\hat{y}_i - y_i$ instead of $y_i - \hat{y}_i$).
- $r_{xy} = R^2$, or the sign of r_{xy} ignored.
- Slope interpreted without units or as a proven causal effect.
- Multiple regression: the clause "holding the other variables fixed" forgotten; "not significant" read as "no effect".

Overview

1. Task 1: Statistical inference: z-test and confidence interval
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3. Task 3: Classification: logistic regression and evaluation

Task 3 – scenario

Logistic regression

$Y = 1$: customer cancels (churn)

x: hotline calls in the last six months

$$\hat{\eta}(x) = -3 + 0.5\,x$$

$$\hat{p}(x) = \frac{e^{\hat{\eta}}}{1 + e^{\hat{\eta}}}$$

Test data, $t = 0.5$

	Pred. No	Pred. Yes	
Act. No	140	20	160
Act. Yes	10	30	40
	150	50	200

Task 3 – the concepts you need

- **Odds:** odds $= \dfrac{p}{1-p}$; back: $p = \dfrac{\text{odds}}{1+\text{odds}}$.
- **Logistic model:** linear predictor $\eta = \beta_0 + \beta_1 x$ (log-odds), $p(x) = \dfrac{e^{\eta}}{1+e^{\eta}}$.
- **Odds ratio** e^{β_1}: factor by which the *odds* change per additional unit of x.
- **Classification rule:** $\hat{y} = 1$ if $\hat{p}(x) > t$; $t = 0.5 \Leftrightarrow \hat{\eta}(x) > 0$.
- **Confusion matrix:** accuracy $= (\text{TP} + \text{TN})/n$, sensitivity $= \text{TP}/(\text{TP} + \text{FN})$, specificity $= \text{TN}/(\text{TN} + \text{FP})$, precision $= \text{TP}/(\text{TP} + \text{FP})$.

Task 3 – from hotline calls to a decision

calls x	$\hat{\eta} = -3 + 0.5x$	odds $= e^{\hat{\eta}}$	$\hat{p} = \frac{\text{odds}}{1+\text{odds}}$	$\hat{y}$ $(t = 0.5)$	$\hat{y}$ $(t = 0.3)$
2	-2.0	0.1353	0.1192	0	0
4	-1.0	0.3679	0.2689	0	0
5	-0.5	0.6065	0.3775	0	**1**
6	0.0	1.0000	0.5000	0	**1**
7	0.5	1.6487	0.6225	**1**	**1**

- Odds of churn overall: $0.2/0.8 = 0.25$.
- One call more: odds $\times\, e^{0.5} = 1.6487$ – the *odds* rise by 65 %, not the probability.
- $t = 0.5 \Leftrightarrow \hat{\eta} > 0 \Leftrightarrow x \geq 7$; $\quad t = 0.3 \Leftrightarrow \hat{\eta} > -0.847 \Leftrightarrow x \geq 5$.

Task 3 – the confusion matrix and its four measures

	Pred. No	Pred. Yes	
Act. No	TN = 140	FP = 20	160
Act. Yes	FN = 10	TP = 30	40
	150	50	200

Measures (which total do we divide by?)

Accuracy	$(30 + 140)/200$	$= 0.85$
Sensitivity	$30/40$ (row Act. Yes)	$= 0.75$
Specificity	$140/160$ (row Act. No)	$= 0.875$
Precision	$30/50$ (column Pred. Yes)	$= 0.6$

Key takeaway

Lower threshold $\Rightarrow$ more customers flagged $\Rightarrow$ sensitivity $\uparrow$, specificity $\downarrow$.

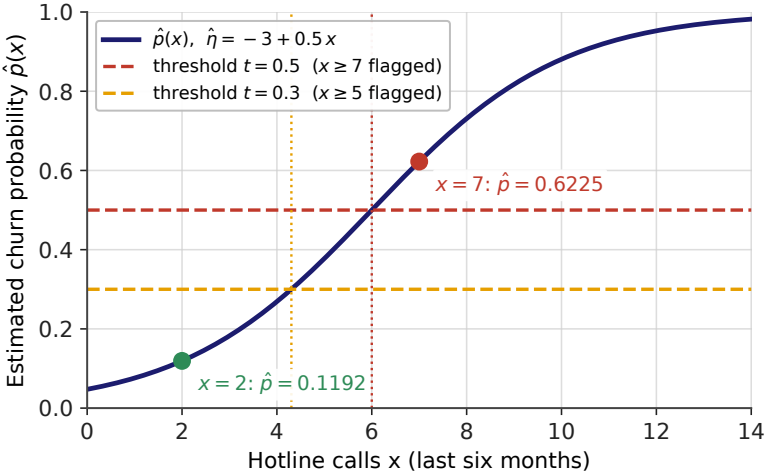
Task 3 – results a)–e)

Part	Asked for	Result
a)	Odds of churn	odds $= 0.2/0.8 = 0.25$
b)	Churn probability at $x = 2$	$\hat{\eta} = -2$, $\hat{p} = 0.1353/1.1353 = 0.1192$
c)	Interpretation of the odds ratio	each extra call multiplies the *odds* of churn by 1.65 (+65 %)
d)	Classification at $x = 7$	$\hat{\eta} = 0.5 > 0$, $\hat{p} = 0.6225 > 0.5 \Rightarrow \hat{y} = 1$ (churn)
e)	Python command (logit)	`smf.logit("churn ~ calls", data=df).fit()`

Task 3 – results f)–j)

Part	Asked for	Result
f)	Accuracy	$(30 + 140)/200 = 0.85$
g)	Sensitivity	$30/(30 + 10) = 0.75$
h)	Specificity	$140/(140 + 20) = 0.875$
i)	Precision	$30/(30 + 20) = 0.6$
j)	Lower threshold $t = 0.3$	sensitivity **increases** (more churners caught)

Task 3 – the S-curve and the two thresholds



Task 3 – typical mistakes

- Reading the log-odds $\hat{\eta}$ (or the odds $e^{\hat{\eta}}$) as a probability.
- "The odds ratio is 1.65, so the *probability* rises by 65 %."
- Mixing up sensitivity (divide by *actual* Yes) and precision (divide by *predicted* Yes).
- Rows and columns of the confusion matrix confused (specificity $\neq \mathrm{TN}/(\mathrm{TN}+\mathrm{FN})$).
- Believing that a lower threshold catches *fewer* churners.

Points and exam strategy

Structure

3 tasks $\times$ 10 subtasks $\times$ 2 points = 60 points in 120 minutes – about 4 minutes per subtask.

- All subtasks are **independent**: skip whatever is stuck and come back to it at the end – you lose no further points.
- Each subtask asks for **one** result. Write it in the answer box.
- Still write down your working: a result without working earns only half the points.
- Do not forget units, signs and rounding.